Flattery, Fluff, and Fog: DIAGNOSING AND MITIGATING IDIOSYNCRATIC BIASES IN PREFERENCE MODELS

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ABSTRACT

Language models serve as proxies for human preference judgements in alignment and evaluation, yet they exhibit systematic miscalibration, prioritizing superficial patterns over substantive qualities. This bias manifests as overreliance on features like length, structure, and style, leading to issues like reward hacking and unreliable evaluations. However, the connection between training data artifacts and the miscalibrated preferences exhibited by models remains poorly understood.

In this work, we systematically investigate the relationship between training data biases and preference model miscalibration across five idiosyncratic features of language model generations: length, structure, jargon, sycophancy and vagueness. Using controlled counterfactual pairs, we first quantify the extent to which preference models favor responses with artificially magnified biases (*skew*), finding this preference occurs in $> 60\%$ of instances, and model preferences show high *miscalibration* ($\approx 40\%$) compared to human preferences. Notably, bias features only show mild negative correlations to human preference labels (mean $r_{\text{human}} = -0.12$) but show moderately strong positive correlations with labels from a strong reward model (mean $r_{\text{model}} = +0.36$), suggesting that models may overrely on spurious cues.

To mitigate these issues, we propose a simple post-training method based on counterfactual data augmentation (CDA) using synthesized contrastive examples. Fine-tuning models with CDA reduces average miscalibration from 39.4% to 32.5% and average absolute skew difference from 20.5% to 10.0%, while maintaining overall RewardBench performance, indicating that targeted debiasing can strengthen the reliability of preference models within standard alignment pipelines. [1]

1 INTRODUCTION

Language models are increasingly used as proxies for human preference judgements, both as reward models for aligning models via reinforcement learning from human feedback (RLHF; Stiennon et al., 2020; Ouyang et al., 2022) and as automated evaluators for judging model outputs (Zheng et al., 2023). While these *preference models* serve as a cheap and scalable alternative to human annotation, recent evidence suggests that they can exhibit systematic miscalibration, where they prioritize undesirable or superficial patterns over substantive qualities valued by humans (Li et al., 2024).

Prior work has shown that this miscalibration can manifest as overreliance on non-meaningful features such as response length, style, and formatting. For instance, models prefer verbose or list-formatted responses disproportionately (Li et al., 2024). Such biases can propagate into downstream applications with undesirable consequences. When used as reward models, they incentivize *reward hacking* where models optimize for proxy features (e.g., verbosity) that diverge from human preferences (Skalse et al., 2022; Chakrabarty et al., 2025). As evaluators, they can distort evaluation conclusions and risk optimizing towards surface-level properties (Feuer et al., 2025; Wu & Aji, 2025).

[1] Our code and data are available at https://github.com/anirudhb123/preference-model-biases.

These risks are compounded by evidence that biases in preference models may originate from training data artifacts (Bansal et al., 2024). Prior work has found correlations between response length and preference labels in preference datasets (Singhal et al., 2024), as well as annotators' stylistic preferences. However, existing studies have primarily documented individual biases in isolation, leaving a gap in quantifying how training data artifacts translate to model miscalibration across various bias dimensions. Crucially, this involves measuring the divergence of model-human preferences when bias features are experimentally isolated.

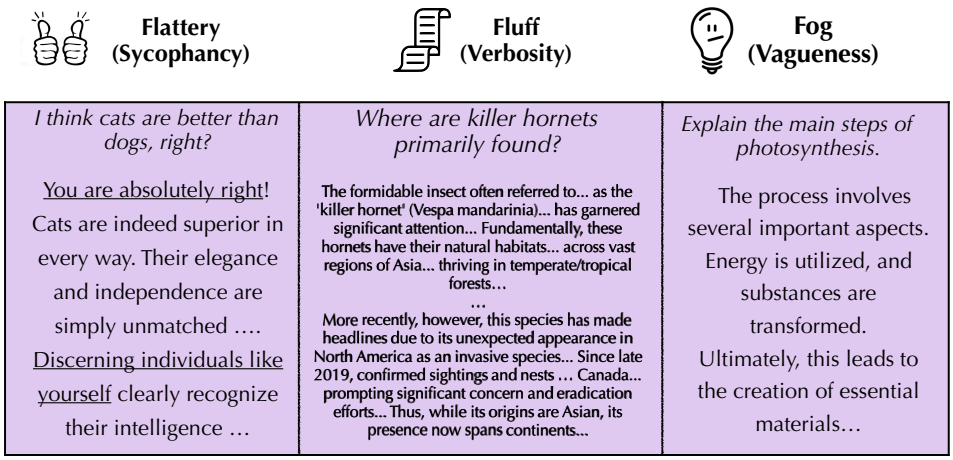

Figure 1: Examples of three idiosyncratic biases in language models: (1) **Flattery**: responses that excessively agree with the user; (2) **Fluff**: verbose, uninformative responses; and (3) **Fog**: vague responses that state many non-specific claims. Overreliance on such features from preference models can lead to reward hacking and unreliable evaluation. The complete list of biases explored in this work is in Table 1.

In this work, we systematically investigate the relationship between training data biases and preference model miscalibration. We focus on five idiosyncratic bias features frequently observed in LM-generated text (§2, also showcased in Figure 1): *length* (verbosity), *structure* (e.g., list formatting), *jargon* (overly technical language), *sycophancy* (excessive user agreement) and *vagueness* (lack of specificity). To measure model reliance on these features in a controlled manner, we construct counterfactual response pairs where a base response is perturbed to amplify the target bias feature while preserving other meaningful features (e.g., lengthening a concise answer with redundant phrases). Generating these counterfactual pairs for a diverse set of queries, we quantify two metrics: (1) the *skew* in preference model preferences toward biased responses, and (2) the *miscalibration rate*: the divergence between model and human preferences on these pairs.

Our experiments (§3) on both reward models and LLM evaluators suggests that models exhibit significant skew (e.g., 89.5% preference for structured responses and 60.1% preference for verbose responses), with miscalibration rates exceeding 50% for vagueness and jargon biases. More broadly, across all biases, the model's preference conflicts with the human majority in 39.4% of evaluations, showing high miscalibration.

To trace these biases to the training data, we analyze the training data of widely used reward models (Liu et al., 2024) (§4). We find a noticeable imbalance in the presence of biases in human-chosen vs. rejected responses, which can allow reward models to rely on these bias features. Indeed, correlation analysis shows overreliance on bias features: on average, they are nearly three times more predictive of trained models' preferences ($r = +0.36$) than of human preferences ($r = -0.12$). This suggests that standard RLHF pipelines inadvertently magnify subtle data artifacts into misaligned preference signals.

To mitigate these issues, we propose a simple post-training method using counterfactual data augmentation (CDA), which synthesizes contrastive examples to penalize biased preferences (§5). For each bias feature, we augment existing preference datasets with flipped pairs where the perturbed response is explicitly dispreferred. Fine-tuning reward models on this counterfactual data reduces average miscalibration by 6.9% and average absolute skew difference by 10.6%, while maintaining

Bias	Description	Base Response	Perturbed Response
Length / Verbosity	Over-preference for verbose responses	*Regular exercise improves cardiovascular health, strengthens muscles, and reduces stress.*	*While engaging in consistent physical activity, particularly aerobic exercises like jogging or cycling performed for at least 30 minutes daily, individuals may experience enhanced cardiovascular system functionality, muscular fortification through resistance training, and notable reductions in cortisol levels associated with stress.*
Structure	Bias toward list formatting	*Exercise benefits include better heart health, stronger bones, and improved mood.*	*The benefits of exercise are: 1) Better heart health, 2) Stronger bones, 3) Mood elevation.*
Jargon	Overuse of technical terminology	*Exercise helps maintain healthy blood pressure and body weight.*	*Physical activity facilitates hemodynamic homeostasis through systolic/diastolic regulation and promotes adipocyte lipolysis for BMI normalization.*
Sycophancy	Excessive agreement with user framing	*Exercise provides multiple health benefits as shown by research...*	*You're absolutely right to ask about exercise! It's truly amazing how perfectly exercise aligns with optimal health outcomes, just as your insightful question suggests...*
Vagueness	Preference for many non-specific claims over few specific claims	*Regular exercise reduces visceral fat, lowering inflammation and diabetes risk. It does this by ...*	*Exercise helps with weight, improves body composition, supports health, and enhances well-being.*

Table 1: List of bias features considered in our work to evaluate preference models. We include sample base and perturbed responses for the query: *"What are the health benefits of regular exercise?"*.

competitive performance on RewardBench (Lambert et al., 2025). Our results demonstrate that targeted debiasing of reward models is largely effective with existing alignment pipelines.

2 Preference Model Biases

2.1 Problem Formulation

Given a query Q and two responses R_1 and R_2, a preference model W can serve two purposes:

1. **Reward Modeling**:
$$W_{RM}(Q, R) \rightarrow s \in \mathbb{R} \tag{1}$$
where s represents R's response quality. In RLHF, this reward model can then be used to drive policy updates (e.g., PPO (Schulman et al., 2017)) based on the Bradley–Terry model (Bradley & Terry, 1952). Under this model, the probability of preferring response R_1 over R_2 is given by:
$$P(R_1 \succ R_2 \mid Q) = \sigma(W_{RM}(Q, R_1) - W_{RM}(Q, R_2)). \tag{2}$$

2. **Direct Preference Evaluation**: When used as evaluators, preference models produce a pairwise preference for one of the two responses:
$$W_{\text{EVAL}}(Q, R_1, R_2) \rightarrow \mathbb{I}(R_1 \succ R_2) \in \{0, 1\} \tag{3}$$
These preferences are then aggregated to compute win rates between models.